Short Analysis Pipeline

BIOLM0051

Archie Benn

Scenario

A suitcase containing several packages of unidentified frozen meat has been intercepted at Heathrow Airport by UK Border Force officers during routine inspections. The shipment had no import documentation, raising concerns that the meat may originate from protected or endangered species.

Given the serious implications for public health, food safety, and wildlife conservation, the case has been referred to the Animal and Plant Health Agency (APHA). As a bioinformatician assisting with the inquiry, you have been tasked with determining the species of origin for each meat sample. DNA barcoding has already been performed in the laboratory, and you have now received the resulting FASTAQ sequence files for analysis.

Abstract

Introduction

Methods

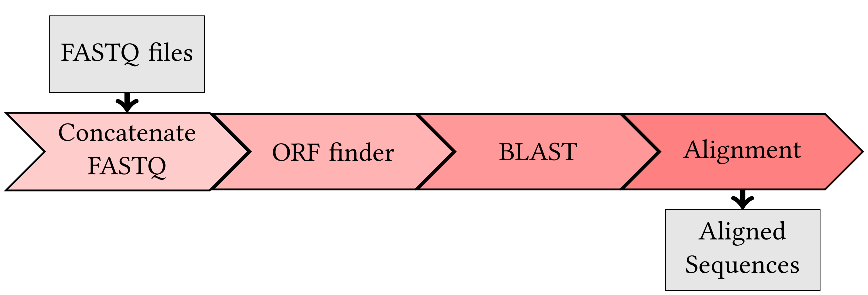

Figure 1: Workflow of analysis pipeline.

Some python code:

```python
from Bio import SeqIO
for record in SeqIO.parse("sequence.fasta", "fasta"):
    print(record.id, record.seq.translate())
```

Some Bash script:

```bash
#!/bin/bash
#SBATCH --job-name=blast_job
#SBATCH --partition=teach_cpu
#SBATCH --time=01:00:00
#SBATCH --mem=2000M

blastn -db nt -query sequence.fa -out results.out -remote
```

Results

In this study, we analysed the gene expression levels across multiple samples using RNA-seq data. The expression values x_i for gene i were normalised using the log-transformation:

$$y_i = \log_2(x_i + 1)$$

to stabilise variance. Data processing was performed using Python scripts:

```python
import pandas as pd
data = pd.read_csv("expression.csv")
normalized = data.apply(lambda x: np.log2(x+1))
```

Alignment results using Clustal Omega on some test data:

```
test1      KMIFVGIKKKEERADLIAYLK   100
test2      KMIFAGIKKKGERADLIAYLK   100
test3      KMAFGGLKKEKDRNDLITYLK   105
consensus  KMIF-GIKKK-ERADLIAYLK

           X  non-conserved
           X  ≥40% conserved
           X  ≥50% conserved
```

Figure 2: Alignment using Clustal Omega and Identity mode in texshade

Discussion

Yadadadada [1] and then also [2] and maybe even from this guy [3, 2]

References

[1] WJ Kent. Blat—the blast-like alignment tool. *Genome research*, 12(4):656–664, 2002.

[2] MJ Wolin, TL Miller, and CS Stewart. Microbe-microbe interactions. In *The rumen microbial ecosystem*, pages 467–491. Springer, 1997.

[3] B Baker, P Zambryski, B Staskawicz, and SP Dinesh-Kumar. Signaling in plant-microbe interactions. *Science*, 276(5313):726–733, 1997.

Supplementary Data

Supplementary data can be found within the GitHub repository for this project: `https://github.com/archiebenn/intro_to_bioinformatics_report.git`